OFFICIAL DEPLOYMENT GUIDE

Pulse Monitoring Tool

A modern, consolidated infrastructure observability platform engineered specifically for virtualized, containerized, and physical compute environments. Consolidates Proxmox VE clusters, Ceph, ZFS, and Proxmox Backup Server into a unified dashboard.

1. What Pulse Monitoring Tool Does

Pulse provides unified, real-time infrastructure visibility without the operational overhead of heavy, complex enterprise agents. It acts as a central aggregator that consolidates virtual machine metrics, storage arrays, backup verification, and physical hardware health into one intuitive pane of glass.

1.1 Core Telemetry & Consolidated Dashboard Scope

Hypervisors & Guests

- Proxmox clusters & individual nodes
- Virtual Machine (VM) and LXC status
- Real-time CPU, RAM, disk, & network use
- Allocated vs. consumed resource analysis
- Node, guest, & capacity alerts

Storage & Backups

- Proxmox storage pool utilization
- Ceph cluster health & OSD status
- ZFS pools, datasets, snapshots, & replication
- PBS datastores, backup jobs, & verifications
- Replication and backup failure tracking

Physical Hardware

- Drive SMART health & wear-out levels
- CPU & system temperature diodes
- Linux software RAID (`mdadm`) state
- Hardware RAID controller diagnostics
- Long-term metric trends & historical forecasting

1.2 Two-Tier Monitoring Architecture: Agentless API vs. Host Agent

Pulse employs a hybrid monitoring model. It prioritizes non-intrusive API connections and reserves agent installation strictly for hardware-level telemetry:

MONITORING TIER	CONNECTION METHOD	WHAT IT MONITORS	WHEN TO USE
Tier 1: Agentless API Token	HTTPS connection (Port 8006) using a scoped Proxmox API Token (`PVEAuditor`).	Cluster inventory, node status, VM & container states, CPU/RAM/Disk consumption, Ceph & ZFS health.	**Recommended default** for all hypervisors. Zero daemon overhead, zero hypervisor risk.
Tier 2: Unified Host Agent	Lightweight system daemon (`pulse-agent`) talking to Pulse over TCP 7655.	Physical drive SMART metrics, CPU/motherboard temperatures, IPMI sensors, `mdadm` RAID, Docker/Podman.	Install on physical nodes where deep hardware diagnostics are required.

1.3 Network Bridge Telemetry Caveat & Recommendation

Important Operational Boundary: Linux Bridges & VLANs

Pulse visualizes overall network throughput for nodes and guests. However, if your operations require fine-grained Linux bridge metrics—such as individual `vmbr0/1/2` throughput, VLAN-tagged bandwidth breakdown, Linux bridge port STP state, packet drops, frame errors, or bond-member health—Pulse should be complemented by **Prometheus + node_exporter + Grafana** or **LibreNMS**.

2. Pulse Server: Prerequisites & Offline Installation

The Pulse monitoring server is deployed directly on a dedicated Linux VM (Ubuntu/Debian) as a native **systemd** service. This approach eliminates the performance overhead, storage abstractions, and port-mapping complexities associated with Docker.

2.1 Target Environment & File Preparation

System Specifications

Target OS	Ubuntu / Debian Linux VM
Service Manager	Native `systemd` (No Docker)
Listening Port	**TCP 7655**
Install Directory	`/opt/pulse-install`
Config Directory	`/etc/pulse`

Required Offline Packages

Due to restricted outbound Internet on VM subnets, place these 3 files in `/opt/pulse-install`:

- `install.sh` (Pulse installer script)
- `pulse-v6.3.1-linux-amd64.tar.gz` (Release package)
- `pulse-v6.3.1-linux-amd64.tar.gz.sshsig` (Signature)

2.2 Step-by-Step Installation Runbook

1 Pre-Installation Validation: Verify Files & Signatures

Confirm files exist, test archive integrity, and ensure the cryptographic signature sidecar is present:

```
cd /opt/pulse-install
ls -lh install.sh pulse-v6.3.1-linux-amd64.tar.gz pulse-v6.3.1-linux-amd64.tar.gz.sshsig

# Validate that signature file is recognized
test -f /opt/pulse-install/pulse-v6.3.1-linux-amd64.tar.gz.sshsig && echo "Signature file found"

# Test tarball integrity and script syntax
tar -tzf /opt/pulse-install/pulse-v6.3.1-linux-amd64.tar.gz | head -10
bash -n /opt/pulse-install/install.sh
```

Installer Archive Rule

The installer's `--archive PATH` flag automatically requires a matching `PATH.sshsig` in the exact same directory. Do not rename or separate these files.

2 Execute Native Local Installation

Run the installer pointing directly to the local archive without redownloading from GitHub:

```
cd /opt/pulse-install
sudo bash ./install.sh --version v6.3.1 --archive /opt/pulse-install/pulse-v6.3.1-linux-amd64.tar.gz
```

3 Verify Service & Listening Port

Ensure the daemon is active, enabled on system reboot, and actively listening on port 7655:

```
# Verify systemd service status
sudo systemctl status pulse --no-pager
sudo systemctl enable --now pulse

# Check TCP port 7655 and local response
sudo ss -lntp | grep 7655
curl -I http://127.0.0.1:7655
```

3. Initial Setup, Security Wizard & Firewall Hardening

After the Pulse service is active, you must retrieve the initial one-time bootstrap token, configure administrative credentials, and permit network access through the host firewall.

4 Retrieve the Initial Bootstrap Token

Pulse uses a one-time bootstrap token to authenticate the administrator on initial setup:

```
# Retrieve the plaintext bootstrap token
sudo /usr/local/bin/pulse bootstrap-token
```

Critical Warning: Do NOT Read .bootstrap_token File Directly

Do **not** attempt to read `/etc/pulse/.bootstrap_token` with `cat`. In Pulse v6, that file contains an encrypted JSON snapshot rather than the usable token. Always run `sudo pulse bootstrap-token` to decrypt and display the valid token.

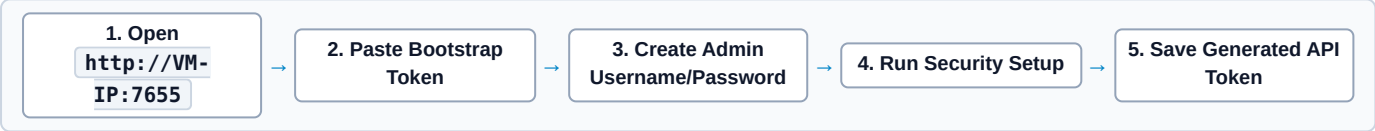
5 Firewall Configuration: Restrict Management Access

Secure port 7655 by allowing only your trusted administration network subnet:

```
# Allow admin subnet (replace 10.31.0.0/23 with your actual management subnet)
sudo ufw allow from 10.31.0.0/23 to any port 7655 proto tcp
sudo ufw status

# Test connectivity from your workstation
nc -vz 10.31.0.50 7655
```

6 Complete Web GUI First-Time Security Wizard



Mandatory Action: Secure the Generated API Token

During the Quick Security Setup wizard, Pulse generates an administrative API token used for automation and agent enrollment. **Copy and securely store this token** before navigating away from the page, as it will not be displayed again.

3.1 Monitored Host Proxy Exclusion (NO_PROXY)

If monitored Linux VMs or Proxmox hosts use an internal proxy (e.g. Squid at `10.31.1.159`), you must explicitly bypass the proxy for Pulse server traffic. Otherwise, agents will fail to connect and display as offline:

```
# Add to /etc/environment or profile on monitored Linux hosts:
export NO_PROXY="localhost,127.0.0.1,10.31.0.50"
export no_proxy="$NO_PROXY"
```

4. Adding Systems to Monitor & Operational Health

Once the Pulse server is initialized, connect your infrastructure via the web dashboard at *Settings → Infrastructure*.

4.1 Method A: Proxmox VE via API Inventory (Agentless)

PVE Settings in Pulse UI

Navigate to: *Settings → Infrastructure → Add Source → Proxmox VE → API Inventory*

Host URL	`https://10.31.x.x:8006`
Token ID Format	`<user>@<realm>!<name>`
Configured ID	`pulse-monitor@pve!pulse`
Permissions	`PVEAuditor` on `/` (Read-Only)
Ignore Self-Signed	**Enabled** (Default certs)

Pre-Flight Token Test via cURL

```
# Validate token from any Linux machine
curl -k \
  -H "Authorization: PVEAPIToken=pulse-
monitor@pve!pulse=SECRET" \
  https://YOUR_PROXMOX_IP:8006/api2/json/version

# Output: { "data": { "version": "..." } }
```

> `PVEAuditor` provides read-only visibility into VMs, storage, and nodes without granting any administrative privileges.

4.2 Method B: Proxmox Backup Server (PBS) Automated Integration

Pulse automates PBS enrollment. In the UI, generate the PBS setup command, review it, and execute it on the PBS host as root:

1. UI Generates Setup Command → **2. Run on PBS as Root** → **3. Creates `pulse-monitor@pbs`** → **4. Grants Read-Only `Audit` Role** → **5. Auto-Registers to Pulse**

4.3 Method C: Standalone Linux Hosts (Unified Agent)

In Pulse, go to *Settings → Infrastructure → Install on a host*. Select Linux and copy the generated command. Run that command on the target VM to automatically enroll the unified agent (`pulse-agent.service`).

4.4 Post-Installation Verification & Maintenance Checklist

```
# Complete Server Health Check Suite:
sudo pulse --version
sudo systemctl is-active pulse
sudo systemctl is-enabled pulse
sudo ss -lntp | grep 7655
sudo journalctl -u pulse -n 20 --no-pager
```

Operational Maintenance Best Practices

- **Backup Config:** Regularly back up `/etc/pulse`.
- **Alerts:** Configure thresholds under *Settings → Alerts*.
- **Updates:** Run `sudo /bin/update` for systemd binary updates.
- **Data Retention:** Set metric history retention in System Settings.